BT1
3S LiPo 11.1V 6Ah
+12V_B
GND
SW1
Main switch
F1
15 A fuse
+12V
U5
MD13S left
M_L_A
M_L_B
J1
Motors JST
M_R_A
M_R_B
U4
INA219
+12VL
SDA
SCL
3V3
U3
Mini560 5V/5A
+5V
U1
Raspberry Pi 5
U2
ESP32-WROOM-32E
BAT_ADC
MTR_ERR
U6
MD13S right
U7
Laser driver + nFET
LASER_P
LASER_N
D1
445 nm 2 W diode
LASER_EN
LASER_ON_HW
SW2
Key lock
U8
HC-SR04 front-left
TRIG_L
ECHO_L
U10
VL53L0X front
XSHUT_L
LS1
Buzzer
BUZZ
U9
HC-SR04 front-right
TRIG_R
ECHO_R
U11
VL53L0X rear
XSHUT_R
D2
Power LED
U12
Servo yaw MG996R
PWM_YAW
U13
Servo shoulder DS3218
PWM_SH
U14
Servo elbow DS3218
PWM_EL
SW3
Emergency stop
ESTOP_N
U15
UBEC 6V 10A
+6VS

Laser safety: Pi EN AND hardware key lock; e-stop kills motors+laser
Drive: 2x MD13S, 8 wheels; Sensors: 2x HC-SR04, 2x VL53L0X (I2C)
3S LiPo -> fuse/switch -> INA219 shunt -> buck 5V -> Pi 5 + ESP32
Semi-Autonomous Weeding Robot

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Title: Weed rover electronics

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